Pre-clinical Operative Dentistry

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Restorative techniques for Cl III

A matrix is applied in Cl III before the insertion of the restorative material to

(1) confine the restoration material excess and

(2) assist in the development of the appropriate axial tooth contours.

The matrix usually is applied and stabilized with a wedge before application of the adhesive because it helps contain the adhesive components to the prepared tooth, it also reduces the amount of excess material, thus minimizing the finishing time.

A wedge is needed at the gingival margin to

(1) hold the Mylar matrix in position,

(2) provide slight separation of the teeth, and

(3) prevent a gingival overhang of the composite material.

A properly contoured and wedged matrix is a prerequisite for a restoration involving the entire proximal contact area, unless the adjacent tooth is missing in which case the restoration may be completed by direct placement of the composite.

A properly contoured section of Mylar matrix material is used for most Class III and IV preparations. Because the proximal surface of a tooth is usually convex inciso-gingivally and the matrix material may be flat, it is necessary to shape the matrix to conform to the desired tooth contour fig1.

The matrix is extended at least 1 mm beyond the prepared gingival and incisal margins. Sometimes, the matrix does not slide through or is distorted by a tight contact or preparation margin.

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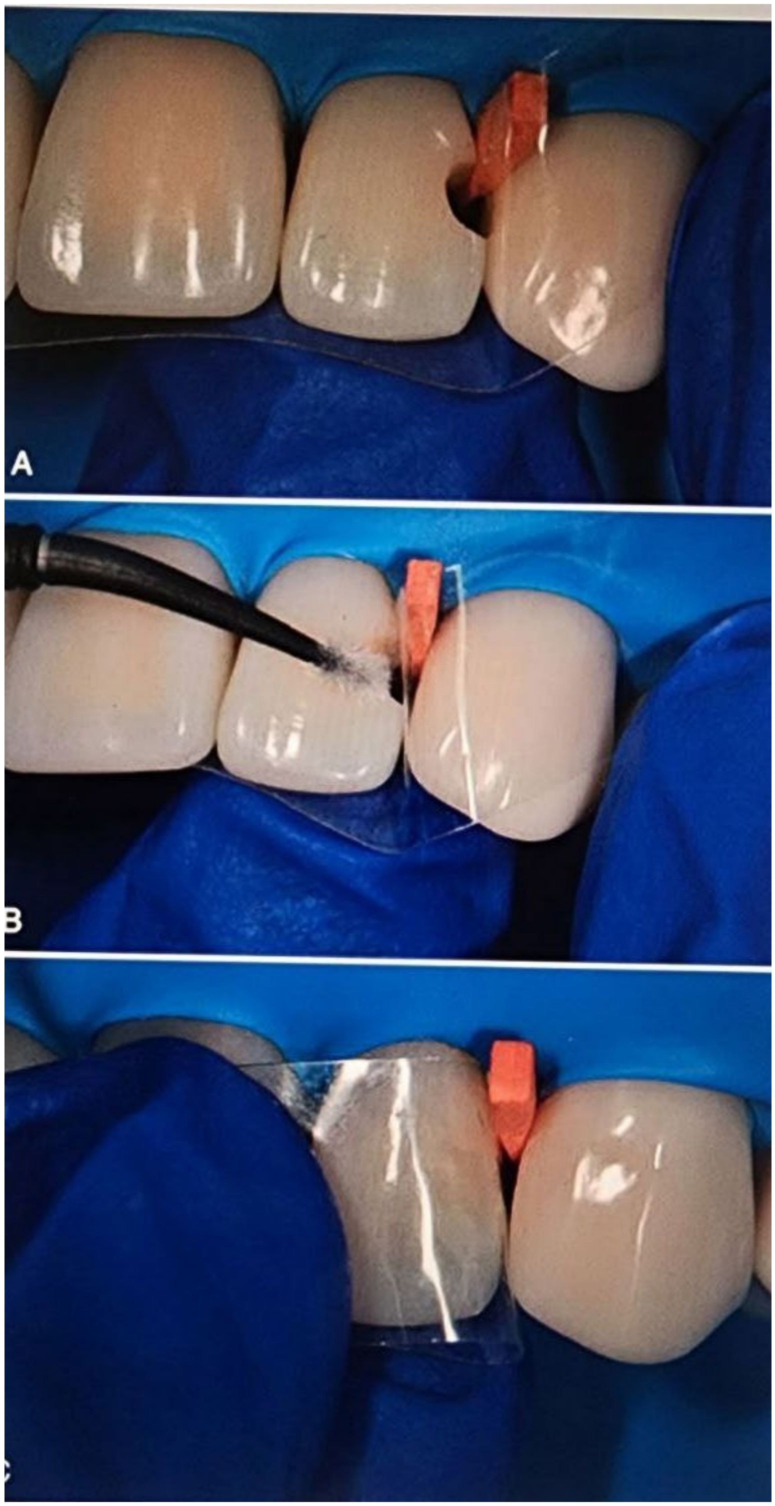

Light-cured materials are not dispensed until ready for use. the composite should be protected from ambient light to prevent premature polymerization. the composite is inserted by a hand instrument and pressed into the preparation.

Most modern composites will not stick to a clean instrument, but if stickiness is perceived, the hand instrument should be thoroughly cleaned and dried, once the desired aspect of the preparation is adequately restored, the material is polymerized. A second increment of composite is applied, if needed, to fill the preparation completely.

Any gross excess is removed quickly with the blade of the insertion instrument or an explorer tine before closing the matrix. the operator closes the lingual end of the matrix over the composite and holds it with the index finger. Next, the operator pulls the matrix toward the facial direction to ensure excellent adaptation of the composite with the facial margin. Before light curing the composite, the operator closes the facial end of the matrix over the tooth with the thumb and index finger of the other hand, tightening the gingival aspect of the matrix ahead of the incisal portion. he closed matrix is manually shifted so as to provide proximal contours that allow normal embrasure form and contact with the adjacent tooth, and the composite is light cured.

Note: If the restoration is under- contoured, more composite can be added to the previously placed composite and light cured. No etching or adhesive is required between layers if the surface has not been contaminated, and the oxygen-inhibited layer remains.

With large restorations, it is better to add and light cure the composite in several increments to reduce the effects of polymerization shrinkage and to ensure complete light curing in remote regions.

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Restorative techniques for CL IV

Mylar matrix, described previously, also can be used for most Class IV preparations, although matrix flexibility makes control difficult this difficulty may result in an over-contoured or under-contoured restoration, open contact, or both. Also, composite material extrudes incisally, but this excess can be easily removed when contouring and finishing.

Creasing (folding) the matrix at the position of the lingual line angle helps reduce the potential under-contouring (rounding) of that area of the restoration. the matrix is positioned and wedged as described for the Class III composite technique.

Gingival overhangs and open contacts are common with any matrix technique

that does not employ gingival wedging, the shade should be selected before isolating the area.

After application of the adhesive, the operator inserts the composite with a hand instrument or a syringe as described earlier for Class III restorations, the composite is inserted in increments less than 2 mm thick. It is usually helpful to develop the lingual surface of the restoration first, then its body, and finally the facial surface.

this approach facilitates the development of adequate anatomy with less potential for resultant excess composite material, this anatomic incremental layering also facilitates the development of adequate shade characterization, as dentin and enamel composite shades can be applied according to the structure they are replacing.

When using a properly contoured Mylar matrix, care must be taken when closing the matrix not to pull with excessive force because the soft material is extruded incisally and results in an under- contoured restoration. When this happens, composite should be added to restore proper contour and contact.

Incremental insertion of light-cured composites also enables the clinician to layer composites with different optical properties (more opaque, darker in color, or both, to mimic dentin; and more translucent, lighter in color, or both, to mimic enamel), which can result in natural-looking composite restorations. After polymerization, the lingual matrix is removed, to ensure optimal polymerization, the operator light cures the restoration from facial and lingual directions.

• Fig. 8.43 Class IV tooth preparation and restoration. A, Extraoral view, minor traumatic fracture. B, Intraoral view. C, Fractured enamel is roughened with a flame-shaped diamond instrument. D, The conservative preparation is etched, while adjacent teeth are protected with Mylar strip. E and F, Contouring and polishing the composite. G, Intraoral view of the completed restoration. H, Extraoral view.

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Finishing and Polishing of the Composite

Good technique in inserting composites significantly reduce the amount of finishing required, usually a slight excess of material will need to be removed so as to provide correct contours similar to tooth preparation rotary instruments, finishing and polishing instruments should be used according to the specific surface being finished and polished. For example, flexible discs and finishing strips are suitable for convex and flat surfaces; oval shaped finishing burs and polishing points are more suitable for concave surfaces; and polishing cups can be used in both convex and concave surfaces.

Restorative techniques for Cl V

The composite can be inserted with a hand instrument or syringe, composites and RMGIs (resin modified glass ionomer) are recommended for Class V restorations.

A light-cured material is recommended for most Class V preparations because of the extended working time and control of contour before polymerization.

Careful placement allows for minimal need of rotary finishing. These features are particularly valuable when restoring large preparations or preparations with margins located on cementum because rotary instrumentation can easily damage the contiguous tooth structure and/or compromise the marginal integrity of the restoration.

After bonding procedures (according to manufacturer’s instructions), it is recommended to insert the composite incrementally with a hand instrument or syringe. the number and position of the increments depend on the size and depth of the preparation.

For large and deep preparations, an incremental technique is recommended.

Deep preparations with retentive undercuts are usually filled with at least two axially placed increments with minimal contact on the occlusal/incisal and gingival walls.

Regardless of the technique used, before light curing the last increment that defines the contour of the restoration, the material is shaped as close to the final contour as possible. An explorer or blade of a composite instrument, is useful in removing excess material from the cervical margin and obtaining the final contour, the light source is applied for polymerization fig 3

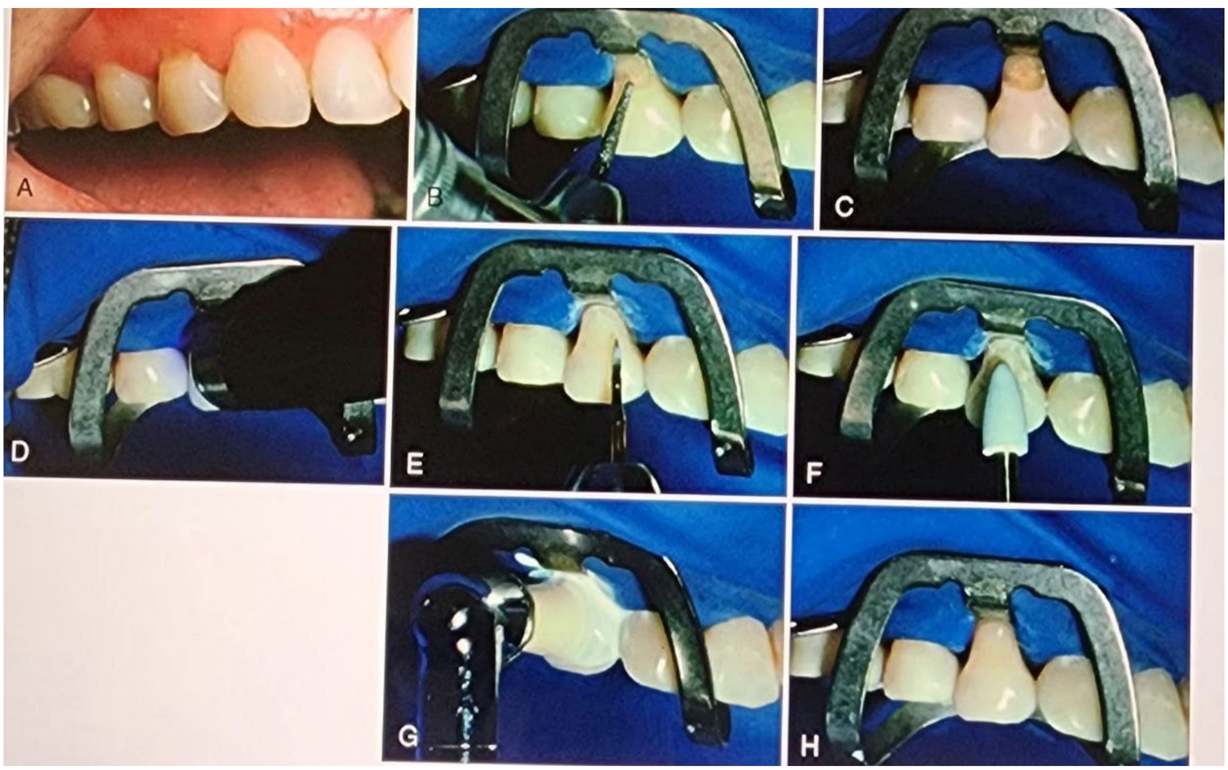

Fig3: Cl V

Finishing and Polishing of the Composite

When excess composite is present, a lame-shaped carbide finishing bur or diamond is recommended for removing excess composite on the facial surface (see Fig. 3 E).

Medium speed with light intermittent brush strokes and an air coolant are used for finishing the restoration to the correct contours. Final polishing is achieved with a rubber polishing point (see Fig. 3 F) or cup, diamond impregnated polisher, and sometimes a polishing paste (see Fig. 3 G and H).

For some locations, abrasive discs, as stated earlier, mounted on an appropriate mandrel in an angled handpiece at low speed can be used.